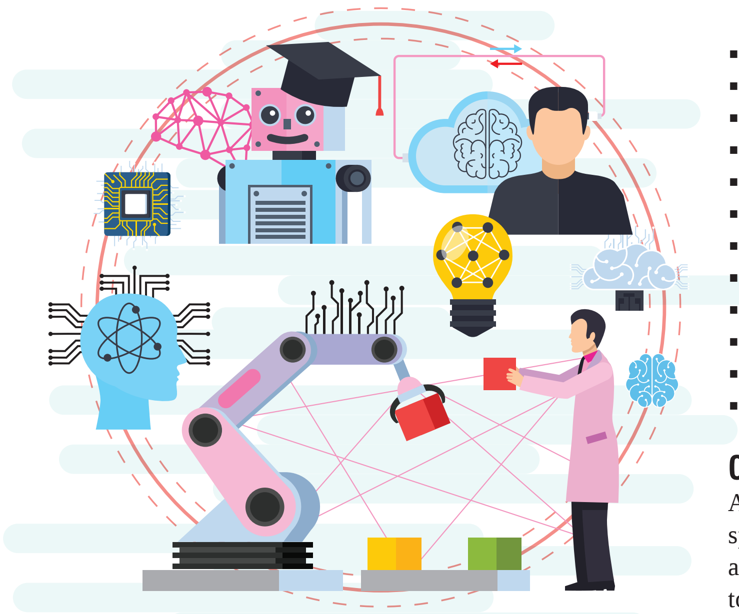
Working with Pre-Trained Deep Learning Models for NLP



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Deep learning is a subset of artificial intelligence that uses artificial neural networks to model and imitate the functioning of the human brain. This article takes a look at how to work with pre-trained deep learning models for natural language processing (NLP).

Deep learning powers numerous artificial intelligence apps and services that automate analytical and physical processes. It is currently used in a very wide range of applications in the government, corporate and social sectors.

A few key applications of deep learning in real-world domains are:

- Real-time computer vision and image analytics
- Virtual assistants
- Automated manufacturing
- Speech recognition (vocal artificial intelligence)
- Data science and engineering
- Entertainment and musical notations
- Stock trading and financial data analytics
- Shopping patterns analysis in e-commerce
- Sentiment analysis on social media
- Customer relationship management systems
- Advertising and promotional activities
- Autonomous vehicles, self-driving cars and drones
- Natural language processing (NLP)
- Fraud detection and cyber security
- Emotional intelligence
- Healthcare and medical diagnosis
- Investment modelling

Classification of models in deep learning

A number of deep learning models are dedicated for a specific application area. In each field of research and application, a particular deep learning model is implemented to enable a higher degree of effectiveness, performance and accuracy (see Table 1).

Table 1: Deep learning models and use cases

Deep learning model	Use cases
Classic neural networks (multilayer perceptrons)	Tabular data analysis, Classification and regression based problem solving
Convolutional neural networks	Image data sets, Optical character recognition (OCR) intelligence
Recurrent neural networks	Image classification, Image captioning, Sentiment analysis, Video classification
Self-organising maps (SOM)	Dimensionality reduction, Music, Video
Auto encoders	Huge data sets, Recommendation engines, Dimensionality reduction
Boltzmann machines	Monitoring and surveillance based applications